

KUMAR GAURAV

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SUMMARY

Software Engineer with strong expertise in **Data Structures & Algorithms, System Design, and Backend Development**. Proven experience in designing and implementing **scalable full-stack applications, RESTful APIs, and low-level networking systems**, along with **database optimization and performance tuning** to deliver reliable, high-impact software solutions.

EDUCATION

Indian Institute of Information Technology and Management, Gwalior
Integrated Postgraduate (B.Tech + M.Tech) in Information Technology

Dec 2021 – May 2026
CGPA: 7.8

EXPERIENCE

LVL ALPHA Pvt Ltd
Full Stack Developer

May 2024 – Jul 2024
Pune, Maharashtra, India

- **Built** a full-stack **Tool Tracking Management System (TTMS)** using **React, Node.js, MongoDB, and Google Maps API**, improving asset visibility for **500+ tools**.
- **Strengthened API security** using **JWT authentication** and **Postman validation**, reducing vulnerabilities by **25%**.
- **Engineered** a **scalable backend architecture**, improving throughput by **29%** under **high-concurrency workloads**.

Research Intern

Aug 2024 – Oct 2024

Hybrid Tri-Band Terahertz Antenna Design

Gwalior, Madhya Pradesh, India

- **Modeled and simulated** a triple-band terahertz microstrip patch antenna with wide and narrow band characteristics, achieving **-47 dB** return loss, **6.8 dBi** gain, and **>92% efficiency**.
- **Optimized** and validated performance using **HFSS and CST Studio** through parametric tuning.

PROJECTS

C++ based Low-Level HTTP Server | C++, TCP/IP, POSIX Sockets, HTTP

Apr 2024 – May 2024

- **Architected** a low-level HTTP server using **TCP/IP sockets**, implementing IPv4 client-server communication with low-level network programming.
- **Implemented** a modular architecture using **polymorphism and inheritance**, ensuring safe memory management and preventing socket leaks and **heap fragmentation**.

Codeforces Database System | SQL, MySQL

May 2024 – Jul 2024

- **Developed** and scaled a **MySQL database** for a competitive programming platform, supporting **10,000+ users**, problem sets, and real-time contest standings.
- **Improved** data integrity and reduced redundancy by **45%** using advanced **normalization (up to BCNF)** and optimized entity relationships.

Pathfinding Visualizer | JavaScript, Graph Algorithms [Live Demo]

May 2025 – Jun 2025

- **Designed** a **pathfinding visualizer** supporting **BFS, DFS, Dijkstra, Greedy Best-First, and A*** algorithms, enabling interactive learning through real-time visualization.
- **Implemented** interactive features supporting **1,000+** cell updates, maze generation, and adjustable animation speeds (**10–200 ms**), improving user engagement and usability.

TECHNICAL SKILLS

- **Coursework:** DSA, Operating Systems, Database Management Systems, Computer Networks
- **Frontend Technologies:** JavaScript (ES6+), React.js, Next.js, HTML5/CSS3, jQuery, DOM Manipulation, Responsive Design
- **Styling & UI:** TailwindCSS, CSS3, Animations, Material-UI, Web UX Patterns, Cross-browser Compatibility
- **Programming Languages:** JavaScript, TypeScript, C/C++, Python, Java, SQL
- **Backend & Databases:** Node.js, Express.js, RESTful APIs, MongoDB, PostgreSQL, MySQL
- **Tools & Platforms:** Git/GitHub, VS Code, React DevTools, Postman, CI/CD, Agile, Linux
- **Core Competencies:** Low-Level Design, Database Optimization, Debugging, System Design

ACHIEVEMENTS

- Selected among the **top 3000 students** nationwide for the **Amazon ML Summer School 2025**.
- **Knight (1941 rating)** on **LeetCode** and **Specialist(1451 rating)** on **Codeforces**.
- Solved **800+** DSA problems across **LeetCode, Codeforces, CodeChef**.
- Secured a **98.5 percentile** in JEE Mains 2021.
- **Qualified GATE 2025** in Computer Science and Engineering (CSE).

POSITION OF RESPONSIBILITY

- Technical Member at **RCoY (Rotaract Club of Youth)**, **IIITM Gwalior**.
- **Problem Setter, Junior Programming Contest II (AASF)**.